THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **Methods of Separation in Everyday Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A mixture of sand and iron in 1:1 by mass weighing 40 g. After using a magnet, the iron obtained should be: 1

(a) 20 g **(b)** 0 g **(c)** 10 g **(d)** 40 g

2. Which is a reason for decanting slowly? 1

(a) to dissolve **(b)** to cool

(c) to avoid disturbing the sediment **(d)** to save time

3. A mixture of water and alcohol cannot be separated by decantation because: 1

(a) water is solid **(b)** alcohol is heavy

(c) alcohol is magnetic **(d)** they mix completely

4. Why would using a very fine sieve not separate a mixture of fine sand and salt? 1

(a) both particles are fine **(b)** sand is soluble

(c) salt is a gas **(d)** sieves are magnetic

5. Why is a funnel with a pointed tip better than a flat one for filtration? 1

(a) it is heavy **(b)** directs the liquid into the container

(c) it is nice **(d)** it is cold

6. The method of spinning milk rapidly to separate cream is called: 1

(a) handpicking **(b)** winnowing **(c)** churning **(d)** sieving

7. A mixture of marbles and sand is best separated by: 1

(a) condensation **(b)** sieving **(c)** evaporation **(d)** decantation

8. Which of the following is false about solutions? 1

(a) solutes pass filter paper

(b) they are uniform

(c) they are clear

(d) they scatter light strongly like muddy water

9. A student pours a mixture of sand and water through filter paper twice. The second filtrate is clearer because: 1

(a) sand dissolves

(b) paper becomes magnetic

(c) water evaporates

(d) fine particles left from the first pass are trapped

10. Which pair can be separated by sieving only if their particle sizes are very different? 1

(a) ink and water **(b)** pebbles and fine sand

(c) sugar and water **(d)** salt and water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a centrifuge used to separate blood components? 2

12. How can you obtain pure sugar from impure sugar? 2

13. Why does a gold-washing (panning) method use water? 2

14. How can you separate water from a mixture of oil and water? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the speed of dissolving depends on particle size, stirring and temperature. 3

16. How would you know if a liquid is pure water or salt water using evaporation? 3

17. What is the difference between distillation and evaporation? 3

OR

Design a fair test to find which of salt and sugar is more soluble in water. 3

18. Explain two ways of getting clean water from a muddy puddle in an emergency. 3

OR

How does a vacuum cleaner separate dust from air? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how the properties of substances are used in each stage of a water-treatment plant. 5

OR

Discuss the dangers of drinking untreated water and the home methods to purify it. 5

20. Analyse how a dairy separates cream from milk and makes butter with the methods of separation. 5

OR

Explain why a mixture of sugar and sand is easier to separate in water than by handpicking. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND